

Polynomials – 3

Symmetric functions, Newton's identities, resultant and discriminant

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Theoretical minimum.

Theorem 1 (fundamental theorem on symmetric polynomials). Every symmetric polynomial in $x_1, \dots, x_n$ over a ring R is a polynomial with coefficients in R in the elementary symmetric polynomials $e_1, \dots, e_n$, and this representation is unique. The proof is an algorithm: kill the leading monomial in lexicographic order by a suitable product of the e_k .

Theorem 2 (Newton's identities). For the power sums $p_k = \sum x_i^k$ one has $p_k - e_1 p_{k-1} + e_2 p_{k-2} - \dots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0$ for $k \leq n$, and the same relation without the last term for $k > n$. Hence p_k is a universal polynomial in the coefficients.

Fact 1 (resultant and discriminant). For $f = a \prod (x - \alpha_i)$ and $g = b \prod (x - \beta_j)$ the resultant is $\text{Res}(f, g) = a^{\deg g} b^{\deg f} \prod_{i,j} (\alpha_i - \beta_j)$; it is a polynomial in the coefficients and vanishes exactly when f and g have a common root. The discriminant is $D(f) = a^{2n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2$, so $D = 0$ exactly when f has a multiple root.

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1. Prove Theorem 1 with the lexicographic algorithm and prove Newton's identities. Then express $p_1, \dots, p_4$ through $e_1, \dots, e_4$ and use the result in the next problem.
 2. (Высшая лига 2016) Let $x_1, \dots, x_5$ be the roots of $F(x) = x^5 + 2x^3 + px^2 + qx + r$ with real p, q, r .
 - a) Prove that F has a non-real root for all real p, q, r .
 - b) Find the value of $x_1^4 + x_2^4 + x_3^4 + x_4^4 + x_5^4$.

3. (Karasev 5.10) Prove that if all roots $x_1, \dots, x_n$ of P are real and simple, then $\sum_{i=1}^n \frac{x_i^k}{P'(x_i)} = 0$ for every $0 \leq k \leq n-2$, and find the value of the sum for $k = n-1$.
4. (S.T. Yau CSMC 2013) Let $p(z)$ have degree $d \geq 2$ with distinct roots $a_1, \dots, a_d$. Show that $\sum_{i=1}^d \frac{1}{p'(a_i)} = 0$.
5. Prove the product formula for the resultant and the criterion for a common root, compute the discriminant of ax^2+bx+c and of x^3+px+q , and use the resultant to eliminate y from the system $x^2 + y^2 = 1$, $x^3 - y = 0$.